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A Multilingual Voice-Enabled Dual-Index RAG Framework for Agricultural Advisory and Government Scheme Assistance

Abstract—Smallholder farmers frequently require two distinct forms of information: practical agricultural guidance for crop and plant problems, and administrative information about government schemes, subsidies, credit facilities, insurance, and eligibility. Conventional large language model (LLM) chatbots can provide fluent responses but may retrieve or generate information outside the intended domain, while language and literacy barriers can limit accessibility. This paper presents a multilingual, voice-enabled, dual-index Retrieval-Augmented Generation (RAG) framework designed to address these issues. The proposed system maintains separate knowledge bases for agricultural diagnostics and government benefit schemes, with independent FAISS dense indices, BM25 sparse retrieval, and SQLite metadata stores. Query processing combines dense semantic retrieval with sparse lexical retrieval, Maximal Marginal Relevance (MMR), and Cross-Encoder reranking. A multilingual pipeline supports English, Hindi, Telugu, and Tamil through script-based language detection, adaptive translation, Faster-Whisper speech recognition, and Edge-TTS speech synthesis. The generation layer is provider-agnostic and supports local and cloud LLM inference. An evaluation framework measures Faithfulness, Context Precision, Context Recall, and Answer Relevance, with baseline configurations for direct LLM, dense-only RAG, and sparse-only RAG. The architecture is intended to reduce domain contamination, improve retrieval precision, and make agricultural information more accessible through text and voice interaction.

Index Terms—Retrieval-Augmented Generation, agricultural AI, multilingual NLP, voice chatbot, hybrid retrieval, BM25, FAISS, Cross-Encoder, government schemes, domain isolation

I. INTRODUCTION

A. Motivation

Agriculture is an information-intensive domain in which timely access to reliable knowledge can influence crop management, disease response, fertilizer use, and access to public support programs. Smallholder farmers may need answers that combine technical agricultural knowledge with administrative information about welfare schemes. However, these two information categories have different semantics and different risks. Crop diagnosis may involve symptoms, treatment practices, pesticides, or fertilizer recommendations, whereas government schemes involve eligibility rules, documentation, benefits, deadlines, and application procedures.

The project addresses this problem through a conversational AI system that combines Retrieval-Augmented Generation with multilingual and voice interaction. The system is explicitly organized around two isolated domains: agricultural

diagnostics and government schemes. This separation is intended to reduce retrieval contamination between technically different knowledge sources while maintaining a common conversational interface.

B. Challenges in Existing RAG Systems

A conventional single-index RAG system can retrieve semantically similar passages from unrelated parts of a corpus. In an agricultural assistant, this may cause a query about crop disease to retrieve government-scheme passages, or a question about subsidy eligibility to retrieve agronomic advice. In addition, dense retrieval may overlook exact policy terms, scheme names, document identifiers, or crop-specific keywords that are well handled by lexical retrieval. Conversely, purely lexical retrieval can miss semantically related passages expressed with different terminology. These observations motivate a hybrid retrieval architecture.

Accessibility presents a second challenge. Farmers may prefer regional languages or spoken interaction rather than English text input. The proposed system therefore incorporates automatic speech recognition, language identification, machine translation fallback, and multilingual speech synthesis.

C. Research Objectives

The main objectives are:

- design a domain-isolated dual-index RAG architecture for agricultural advisory and government benefit navigation;
- combine dense semantic retrieval and sparse BM25 retrieval to improve candidate recall;
- apply Cross-Encoder reranking to improve the ordering of retrieved evidence;
- support English, Hindi, Telugu, and Tamil through adaptive multilingual query processing;
- provide voice input and voice output to improve accessibility; and
- establish an evaluation framework for faithfulness, context quality, answer relevance, recall, and system latency.

D. Contributions

The principal contributions are:

- an isolated dual knowledge-base architecture with independent dense/sparse indices and metadata stores;
- a hybrid BM25–FAISS retrieval strategy with score normalization and MMR diversity ranking;

- Cross-Encoder reranking using BAAI/bge-reranker-base and weighted score fusion;
- an adaptive multilingual query-routing pipeline with translation caching;
- a voice-enabled interface using Faster-Whisper ASR and Edge-TTS; and
- a reproducible evaluation framework with direct-LLM, dense-RAG, and sparse-RAG baselines.

II. RELATED WORK

Retrieval-Augmented Generation combines information retrieval with generative language models so that responses can be conditioned on external evidence rather than relying exclusively on parametric model knowledge. Dense retrieval represents queries and documents in continuous vector spaces and is effective for semantic matching. Sparse methods such as BM25 emphasize exact lexical overlap and remain valuable for specialized terminology, names, identifiers, and policy language. Hybrid retrieval combines these complementary signals.

Reranking provides a further refinement stage. A Cross-Encoder jointly processes a query and candidate document, allowing richer pairwise relevance estimation than independent embedding similarity. In multilingual conversational systems, speech recognition, language identification, machine translation, and speech synthesis can form an accessibility layer over a retrieval and generation backend.

The proposed work combines these established components with domain isolation. Its research focus is the system-level integration of domain-isolated retrieval, hybrid ranking, multilingual routing, voice interaction, and evaluation for agricultural advisory and benefit navigation.

III. PROBLEM FORMULATION

Let Q denote a user query and let $M \in \{A, G\}$ represent the selected domain, where A is agricultural diagnostics and G is government schemes. Each domain maintains its own corpus D_M , dense index F_M , sparse index B_M , and metadata store S_M . Retrieval is restricted to the selected domain:

$$R(Q, M) = \text{Retrieve}(Q, F_M, B_M). \quad (1)$$

The retrieved candidate set is reranked using a Cross-Encoder. The resulting evidence is supplied to an LLM through a grounded prompt. For multilingual input, the system first identifies the language and either retrieves directly from native-language documents or translates the query to English when native documents are unavailable.

IV. PROPOSED SYSTEM METHODOLOGY

A. Overall Architecture

The system follows a layered architecture. A farmer or extension worker interacts with a Streamlit interface through text or microphone input. Voice input is transcribed using Faster-Whisper. Language detection then determines the processing route. The selected domain routes the query to either the agricultural or government-scheme knowledge base. Each

TABLE I
MAJOR SYSTEM LAYERS

Layer	Components	Purpose
Interface	Streamlit	Text/voice interaction and display
API	FastAPI	Asynchronous REST backend
Language	Script detection, translation, Faster-Whisper, Edge-TTS	Multilingual voice pipeline
Domain	Agriculture / Government Schemes	Knowledge-base isolation
Retrieval	FAISS + BM25 + MMR	Candidate retrieval and diversity
Reranking	BGE reranker	Fine-grained relevance ordering
Generation	Ollama / Groq	Grounded response generation
Storage	SQLite + FAISS	Metadata, embeddings, indices, caches

knowledge base contains a FAISS dense index, a SimpleBM25 sparse index, and SQLite metadata. Hybrid retrieval produces candidates, MMR reduces redundancy, and a Cross-Encoder reranks the candidates. A prompt builder constrains the LLM to use retrieved evidence. Responses are cached and may be synthesized into speech.

B. Document Ingestion and Chunking

Documents are parsed from multiple formats, metadata is extracted, and text is recursively segmented using hierarchical separators. The configured target chunk length is 1000 characters with an overlap of 200 characters:

$$\text{Chunk}_i = D[s_i : s_i + L], \quad s_{i+1} = s_i + L - O, \quad (2)$$

where $L = 1000$ and $O = 200$. The overlap preserves continuity across chunk boundaries and reduces the risk that a relevant statement is separated from its supporting context.

C. Dense Retrieval

The dense retrieval component uses BAAI/bge-small-en-v1.5 to represent queries and document chunks in a 384-dimensional embedding space. FAISS is used for efficient nearest-neighbor search. Cosine similarity is computed as

$$S_{\text{dense}}(Q, D) = \frac{\mathbf{v}_q \cdot \mathbf{v}_d}{\|\mathbf{v}_q\|_2 \|\mathbf{v}_d\|_2}. \quad (3)$$

D. Sparse BM25 Retrieval

The sparse component uses a pure-Python SimpleBM25 implementation. Okapi BM25 scores documents according to term frequency, inverse document frequency, and document-length normalization:

$$S_{\text{BM25}}(Q, D) = \sum_i \text{IDF}(q_i) \times \frac{f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1(1 - b + b|D|/\text{avgdl})}. \quad (4)$$

with $k_1 = 1.5$ and $b = 0.75$.

E. Hybrid Retrieval

Because BM25 scores are not naturally bounded in the same way as cosine similarity, the BM25 score is normalized against the maximum score among candidate documents. The final hybrid score is

$$S_{\text{hybrid}} = \alpha S_{\text{dense}} + (1 - \alpha) \tilde{S}_{\text{BM25}}, \quad (5)$$

where $\alpha = 0.5$ and

$$\tilde{S}_{\text{BM25}}(D) = \frac{S_{\text{BM25}}(D)}{\max_{d' \in C} S_{\text{BM25}}(d')}. \quad (6)$$

F. Maximal Marginal Relevance

MMR is used to reduce redundant retrieval:

$$\text{MMR} = \arg \max_{D_i} \left[\lambda S(Q, D_i) - (1 - \lambda) \max_{D_j \in S} S(D_i, D_j) \right], \quad (7)$$

where the configured diversity coefficient is $\lambda = 0.5$.

G. Cross-Encoder Reranking

The top hybrid candidates are passed to BAAI/bge-reranker-base. The raw reranking logit is converted through a sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}. \quad (8)$$

The final score is

$$S_{\text{combined}} = 0.4 S_{\text{hybrid}} + 0.6 \sigma(z_{\text{rerank}}). \quad (9)$$

This weighted fusion gives the Cross-Encoder a greater contribution to final ordering while retaining information from the initial hybrid retrieval stage.

H. Grounded Generation and Caching

The prompt builder supplies selected evidence and imposes grounding constraints and multilingual response requirements. The RAG chain coordinates conversation history, retrieval, reranking, prompt construction, response caching, confidence estimation, and LLM invocation. A sliding conversation window of five turns provides conversational context without allowing unbounded history growth.

V. MULTILINGUAL AND VOICE PIPELINE

A. Speech Recognition

Voice input is processed with Faster-Whisper. The implementation supports int8 execution on CPU/CUDA and is configured for speech recognition with automatic language detection. The implementation targets low-latency processing; the target reported by the project should be verified experimentally on the final deployment hardware.

B. Language Detection and Translation

The language helper uses Unicode script ranges for rapid identification of Devanagari, Telugu, and Tamil text. English queries can be searched directly. For non-English queries, the system checks whether native-language documents are available. If not, the query is translated to English before retrieval. Translation results are cached in SQLite to reduce repeated translation overhead.

TABLE II
IMPLEMENTATION STACK

Module	Implementation
Frontend	Streamlit
Backend	FastAPI + Uvicorn
Dense embeddings	BAAI/bge-small-en-v1.5
Vector search	FAISS
Sparse search	SimpleBM25
Reranking	BAAI/bge-reranker-base
Local LLM	Ollama with Qwen2.5 7B / Llama 3.2
Cloud LLM	Groq with supported Llama/Mixtral models
ASR	Faster-Whisper
TTS	Edge-TTS; gTTS fallback
Metadata/cache	SQLite
Languages	English, Hindi, Telugu, Tamil

C. Speech Synthesis

Edge-TTS is used for neural speech synthesis. The configured voices include en-US-GuyNeural for English, hi-IN-MadhurNeural for Hindi, te-IN-MohanNeural for Telugu, and ta-IN-ValluvarNeural for Tamil.

VI. GOVERNMENT SCHEME KNOWLEDGE BASE

The government-scheme module is a separate knowledge domain rather than an additional collection mixed into the agricultural diagnostic index. It is intended to support questions about schemes, subsidies, eligibility, benefits, documentation, and application-related guidance. The separate ingestion pipeline and storage structure allow the system to route scheme questions only to the scheme corpus while preserving the agricultural diagnostic corpus for plant and crop queries.

This separation is important because the two domains contain different types of entities and decision criteria. A scheme query may depend on farmer category, location, eligibility conditions, or required documents, whereas a crop query may depend on symptoms, disease indicators, or management recommendations.

VII. SYSTEM IMPLEMENTATION

The backend exposes query, voice-query, history, document, and health endpoints. The LLM layer uses an abstract client interface and factory pattern, allowing local and cloud inference providers to be switched without changing the RAG orchestration layer.

The architecture also accounts for constrained deployment environments. Heavy neural models can create memory pressure on low-memory cloud instances; therefore, model loading, embedding generation, and remote inference should be evaluated separately when deploying to resource-limited infrastructure.

VIII. EXPERIMENTAL SETUP

The research evaluation should be conducted using a benchmark set containing representative questions from both domains. The agricultural portion should include crop symptoms, disease identification, treatment guidance, fertilizer and management questions. The government portion should include

TABLE III
BASELINE CONFIGURATIONS

Configuration	Description
Direct LLM	LLM answers without retrieval evidence
Dense RAG	FAISS-based semantic retrieval
Sparse RAG	BM25-only retrieval
Hybrid RAG	FAISS + BM25 score fusion
Hybrid + Reranker	Hybrid retrieval followed by Cross-Encoder
Proposed Dual-Index RAG	Domain isolation + hybrid retrieval + reranking + multilingual routing

scheme eligibility, benefits, documents, application procedures, and related administrative questions. Queries should be distributed across English, Hindi, Telugu, and Tamil where appropriate.

A. Baselines

B. Evaluation Metrics

The implemented evaluator measures four semantic RAG metrics. Faithfulness estimates whether generated answer sentences are supported by retrieved context. Context Precision measures semantic relevance of retrieved chunks to the query. Answer Relevance measures similarity between the query and generated response. Context Recall measures whether reference information is represented in the retrieved context.

The current implementation uses a semantic similarity threshold of 0.65 for sentence-level faithfulness and recall decisions:

$$\text{Faithfulness} = \frac{1}{|S_A|} \sum_s I \left(\max_c \text{Sim}(s, c) \geq 0.65 \right). \quad (10)$$

$$\text{ContextPrecision} = \frac{1}{k} \sum_i \text{Sim}(Q, C_i). \quad (11)$$

$$\text{AnswerRelevance} = \text{Sim}(\text{Embed}(Q), \text{Embed}(R)). \quad (12)$$

$$\text{ContextRecall} = \frac{1}{|G|} \sum_g I \left(\max_c \text{Sim}(g, c) \geq 0.65 \right). \quad (13)$$

C. Additional System Metrics

Additional measurements should include retrieval latency, reranking latency, LLM generation latency, end-to-end latency, ASR latency, translation latency and cache-hit rate, retrieval Precision@K/Recall@K or MRR where ground-truth relevance labels are available, and cross-domain retrieval contamination rate.

IX. RESULTS AND DISCUSSION

The numerical values below are retained from the supplied project benchmark tables. They should be treated as experimental results only if they correspond to an actual executed benchmark run with the stated configuration and dataset.

A. Retrieval Component Ablation Study

B. Multilingual Evaluation

C. Discussion

The primary research hypothesis is that domain isolation can reduce irrelevant cross-domain evidence, while hybrid retrieval can improve robustness to both semantic variation and exact terminology. The reranker is expected to improve final candidate ordering at additional computational cost. Multilingual translation fallback increases language coverage when native documents are unavailable, but introduces an additional translation stage that should be reflected in latency measurements. Voice interaction improves accessibility but introduces ASR and TTS overhead. These hypotheses should be confirmed or rejected using measured results rather than assumed.

D. Domain Contamination Analysis

A particularly relevant experiment is to construct mixed-domain queries and measure how often the retrieved top- k evidence contains chunks from the wrong domain. This directly tests the central architectural motivation.

X. LIMITATIONS

The quality of answers depends on the completeness, correctness, freshness, and language coverage of the underlying documents. Translation fallback may introduce semantic errors, especially for agricultural terminology and policy-specific phrases. The current semantic evaluation metrics depend on embedding similarity thresholds and therefore do not replace expert human evaluation. Cross-Encoder reranking improves relevance estimation but increases computational cost and latency. Cloud deployment on low-memory instances may require remote embeddings, lazy model loading, or higher-resource infrastructure. Government scheme information can change over time; production deployment requires systematic document versioning and update procedures. The architecture should not be described as eliminating hallucination; this must be evaluated empirically.

XI. CONCLUSION AND FUTURE WORK

This paper presented a dual-index, multilingual, voice-enabled RAG architecture for agricultural advisory and government benefit navigation. The proposed system separates agricultural diagnostics from government schemes, combines dense FAISS retrieval with sparse BM25 retrieval, applies MMR for diversity and Cross-Encoder reranking for relevance refinement, and integrates multilingual voice processing. The modular implementation supports both local and cloud LLM inference and includes an evaluation framework for quantitative validation.

Future work includes larger multilingual benchmark datasets, expert evaluation by agricultural specialists, policy-document freshness monitoring, improved cross-lingual retrieval, domain-specific embedding models, stronger eligibility reasoning for government schemes, on-device inference, and systematic latency/memory optimization for edge

TABLE IV
QUANTITATIVE PERFORMANCE BENCHMARKS ACROSS RAG SYSTEM ARCHITECTURES

System Architecture	Faithfulness (%)	Context Precision (%)	Context Recall (%)	Answer Relevance (%)	Latency (ms)
Direct LLM (No RAG)	22.4	–	–	58.2	1850
Sparse BM25 RAG	71.5	61.8	78.5	69.4	320
Dense FAISS RAG	81.2	68.4	85.0	76.8	380
Hybrid RAG (FAISS + BM25)	88.6	72.8	92.5	81.5	410
Hybrid + Cross-Encoder Reranker	94.2	75.4	100.0	86.8	680
Proposed Dual-Index RAG	96.5	78.2	100.0	89.5	710

TABLE V
RETRIEVAL COMPONENT ABLATION STUDY

Component Variant	Research Question	Empirical Finding and Measured Result
Dense Only	How well does semantic retrieval perform alone?	Achieves 68.4% Precision and 85.0% Recall. Strong on concept matching, but misses specific crop disease names and numerical fertilizer ratios.
BM25 Only	How well does lexical matching perform alone?	Achieves 61.8% Precision and 78.5% Recall. Captures exact policy keywords, but fails on synonymous phrasing.
Dense + BM25 (Hybrid)	Does score fusion improve candidate quality?	Boosts Context Recall to 92.5% and Precision to 72.8%. Candidate overlap between Dense and Sparse is only 11.1%, indicating strong complementarity.
Hybrid + MMR ($\lambda = 0.5$)	Does diversity ranking reduce redundant context?	Reduces context sentence redundancy by 42.6%, optimizing token efficiency without loss in recall.
Hybrid + Reranker	Does pairwise reranking improve top- k ordering?	Achieves 100.0% Context Recall and increases Faithfulness to 94.2% by placing the most relevant diagnostic chunk at Top 1 ($S_{combined} = 0.9179$).
Dual-Index Isolation	Does domain separation reduce cross-domain retrieval?	Reported cross-domain wrong-domain retrieval decreases from 31.4% to 0.0%.

TABLE VI
MULTILINGUAL AND VOICE PIPELINE BENCHMARK

Language	Text Precision / Recall	Voice Precision / Recall	ASR WER (%)	Latency (ms)
English	78.2 / 100.0	76.5 / 98.0	4.2	710
Hindi	74.8 / 95.5	72.1 / 93.0	7.8	1120
Telugu	72.5 / 94.0	69.8 / 91.5	9.4	1180
Tamil	71.9 / 93.5	68.9 / 90.5	9.8	1210

deployment. A further research direction is an explicit domain-contamination benchmark comparing mixed-index and domain-isolated retrieval under controlled query sets.

XII. REPRODUCIBILITY

The project can be executed locally using:

```
uvicorn backend.api:app --host 0.0.0.0
--port 8000 --reload
streamlit run app.py
python -m evaluation.benchmark_runner
```

The implementation includes dedicated ingestion pipelines for agricultural documents and government schemes, separate re-

trieval storage, evaluation scripts, and baseline configurations, enabling controlled experiments and reproducibility.

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TABLE VII
CROSS-DOMAIN RETRIEVAL CONTAMINATION BENCHMARK

Domain Query Set	Single Index	Mixed Index	In-Index	Proposed Index	Dual-Index
Crop Diagnostic Queries	28.5%			0.0%	
Government Scheme Queries	34.2%			0.0%	
Overall Contamination Rate	31.4%			0.0%	